DCS HW03 Code

113511150 許邑帆

大綱

- Design
- Optimization
  - Area
  - Latency
- Report

Design

- 分派、狀態機分開設計
  - 先用簡單分配驗證狀態機可行

```
Elevator_FSM E0(
    .clk(clk),
    .rst_n(rst_n),

    .car_req(car_req[7:0]),
    .up_req({up_req[7:5],5'd0}),
    .down_req({down_req[7:5],5'd0}),
    .btn_open(btn_open[0]),
    .btn_close(btn_close[0]),
    .pax_cnt(pax_cnt[3:0]),

    .cur_floor(cur_floor[2:0]),
    .lift_status(lift_status[2:0]),
    .serve_dir(serve_dir[1:0]),
    .is_full(is_full[0])

);
```

Design

- 分派、狀態機分開設計
  - 讓AI幫我寫分派規則，驗證想法
  - 但後來沒有用

```
for (int i = 0; i < 8; i++) begin
    // --------------------------------------------------
    // 處理 UP 需求
    // --------------------------------------------------
    if (up_req[i]) begin
        // 規則 1: Same Stop (看哪台車廂裡面已經有人要去這樓)
        if      (car_req[i])      up_asg0[i] = 1'b1;
        else if (car_req[i+8])    up_asg1[i] = 1'b1;
        else if (car_req[i+16])   up_asg2[i] = 1'b1;

        // 規則 2: 同一路徑 (同方向 UP，且電梯還沒超過該樓層)
        else if (dir0 == DIR_UP && flr0 <= i) up_asg0[i] = 1'b1;
        else if (dir1 == DIR_UP && flr1 <= i) up_asg1[i] = 1'b1;
        else if (dir2 == DIR_UP && flr2 <= i) up_asg2[i] = 1'b1;

        // 規則 3: 分區 Zoning (預設分派)
        else if (i >= 5 && i <= 7) up_asg0[i] = 1'b1; // E0 負責 7~5
        else if (i >= 2 && i <= 4) up_asg1[i] = 1'b1; // E1 負責 4~2
        else                       up_asg2[i] = 1'b1; // E2 負責 1~0 (若真的是E0負責，把這行改成 up_asg0[i] = 1'b1)
    end
```

Design

- 設計要注意的點：一個樓層要停超過10個cycle

```
always_comb begin : floor
    if(serve_dir_next[0] && (lift_status_next == MOVING && lift_status!=IDLE) && cnt_next == 0)
        cur_floor_next = cur_floor + 1;
    else if(serve_dir_next[1] && lift_status_next == MOVING && lift_status!=IDLE && cnt_next == 0)
        cur_floor_next = cur_floor - 1;
    else
        cur_floor_next = cur_floor;
end
```

Optimaztion

Latency

- MOVING可以直接接OPENING

```
MOVING:begin
    if(need_stop && cnt==9)
        lift_status_next = OPENING;
    else
        lift_status_next = MOVING;
end
```

```
MOVING:begin
    if(need_stop)
        lift_status_next = OPENING;
    else
        lift_status_next = MOVING;
end
```

Optimaztion

AREA

- 不分配，三台電梯一直轉，順路的req才開門
- 不分配，三台電梯一直轉，每層樓都開門
- 不分配，兩台電梯一直轉，每層樓都開門

Optimaztion

Area

```
always_comb begin : STATE
    priority case (lift_status)
        IDLE:begin
            lift_status_next = MOVING;
        end
        MOVING:begin
            lift_status_next = OPENING;
        end
        OPENING:begin
            if(cnt[2])
                lift_status_next = OPEN;
            else
                lift_status_next = OPENING;
        end
```

```
always_comb begin : dir
    priority case (lift_status)
        IDLE:begin
            serve_dir_next = UP;
        end
        OPEN:begin
            if( cur_floor == 0)
                serve_dir_next = UP;
            else if( cur_floor == 7 )
                serve_dir_next = DOWN;
            else
                serve_dir_next = serve_dir;
        end
        default: serve_dir_next = serve_dir;
    endcase
end
```

Report

```
19:24 es26dcs108@ee22[~/HW03/01_RTL]$ ./08_check_design
--- 01_RTL PATTERN PASS !! ---
Execution cycles: 1267447 cycles
Cycle Time: 10.0 ns
```

```
Number of ports:                           87
Number of nets:                           177
Number of cells:                          163
Number of combinational cells:            126
Number of sequential cells:                28
Number of macros/black boxes:               0
Number of buf/inv:                         30
Number of references:                      31

Combinational area:               1899.374430
Buf/Inv area:                      302.702411
Noncombinational area:            1776.297615
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (No wire load specified)

Total cell area:                  3675.672045
Total area:                 undefined
```

```
Point                                     Incr       Path
-----------------------------------------------------------
clock clk (rise edge)                     0.00       0.00
clock network delay (ideal)               0.00       0.00
input external delay                      5.00       5.00 f
btn_close[0] (in)                         0.00       5.00 f
U173/Y (AOI21XL)                          0.33       5.33 r
U187/Y (AOI2BB2XL)                        0.22       5.55 r
U186/Y (OAI22XL)                          0.21       5.77 f
U134/Y (NOR4X1)                           0.58       6.35 r
U253/Y (INVXL)                            0.06       6.41 f
U171/Y (AND2XL)                           0.25       6.66 f
U179/Y (AOI21XL)                          0.16       6.82 r
U229/Y (AOI32XL)                          0.18       7.00 f
E0_cur_floor_reg_2_/D (DFFSHQXL)          0.00       7.00 f
data arrival time                                    7.00

clock clk (rise edge)                    10.00      10.00
clock network delay (ideal)               0.00      10.00
clock uncertainty                        -0.10       9.90
E0_cur_floor_reg_2_/CK (DFFSHQXL)         0.00       9.90 r
library setup time                       -0.37       9.53
data required time                                   9.53
-----------------------------------------------------------
data required time                                   9.53
data arrival time                                   -7.00
-----------------------------------------------------------
slack (MET)                                          2.53
```

謝謝！

謝謝跟我分享想法的陳羿嘉、楊承祐、何穎達、陳俊霖